

Dongho Yee

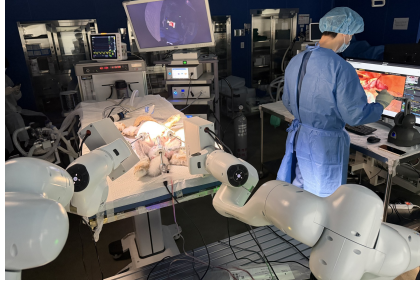
Co-founder & CTO, Rosota Inc. · Researcher, Seoul National University Hospital · B.S. candidate in Mechanical Engineering & Computer Science, Seoul National University

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Dual citizen of the United States and the Republic of Korea · born 2003 in Indiana, USA · based in Seoul



I am a mechanical engineering and computer science student at Seoul National University and the co-founder and CTO of Rosota, a surgical-robotics spin-off of Seoul National University Hospital. I work on the instruments, data pipelines and learning systems that let surgical robots learn from surgeons' own hands. This year I led Rosota's first study: a policy trained only on recordings from surgeons' hand-held instruments completed appendectomies in three of four live rabbits. Before this I ran a small AI startup for three years, worked on GPU code for diffusion models at FriendliAI, and served eighteen months as an attack-helicopter mechanic.



Two Franka FR3 arms under policy control during a live-rabbit appendectomy; the supervising console is at right.



SurgiLogger, a clip-on recorder for standard laparoscopic instruments.



Endoscope view during a rollout: the policy has closed the grasper on the appendix and lifts it toward the dissector.

RESEARCH AND WORK EXPERIENCE

Co-founder & Chief Technology Officer

Mar 2026 – present

Rosota Inc., Seoul — surgical-robotics spin-off of the SNUH MEDISC Lab

Seed round led by FuturePlay with Schmidt and ZDVC (US\$1.1M, Jun 2026); TIPS R&D grant (US\$0.7M, Oct 2026). I lead the device, instrument and surgical-robot (control & policy) teams.

- **SurgiLogger.** Designed the device end to end (mechanics, nRF52840 firmware, BLE link, calibration). It clips onto the shaft of a standard instrument and recovers roll, depression, insertion depth and jaw opening from an IMU, a time-of-flight sensor and a Hall sensor, with nothing added to the part that enters the patient: 0.17–0.42° attitude and 1.36 mm depth RMS against a Franka FR3 ground truth, 20 h per charge, worn by surgeons in the operating room. The current version adds a strain gauge that senses the interaction force between the tip and tissue.
- **Policy and platform.** Built the imitation-learning stack (diffusion policy, DINOv3 ViT-B/16, 16-step action chunks, FiLM phase conditioning) and the dual-FR3 laparoscopic platform with a 1 kHz remote-center-of-motion controller. Chose the design by closed-loop rollouts of 31 configurations in an Isaac Sim scene reconstructed from depth maps of real organs; a frozen self-supervised ViT cut trocar violations from 88% to 6% relative to ResNet-18.
- **In-vivo study (first author).** 849 demonstrations from four live rabbits; the retrained policy completed the appendectomy with electrosurgery in three of four further animals under shared autonomy, peak RCM error 0.84 mm, all animals healthy at seven days. To our knowledge the first learned policy to fire electrosurgery in a live animal. Demonstration data released.
- **UWB tool-tip localization (co-first author).** In the operating room an instrument's absolute position cannot be read from its own sensors: the trocar's location is unknown to them and the sterile drape blocks every optical path, which leaves time-of-flight ranging that scatters by centimeters. I built a ranging sensor accurate to a millimeter instead: clip-on ultra-wideband nodes that range by carrier phase through the drape ($\sigma = 0.10\text{--}0.33\text{ mm}$ at 35–46 Hz), and a trocar-constrained solve that turns three distances into both tool tips in absolute, endoscope-frame coordinates.

In progress

- First human operating-room data collection at SNUH, planned for 20 Nov 2026 (IRB under review, PI Prof. Kyu Eun Lee).
- Force-aware policies from SurgiLogger's strain-gauge channel, and retargeting 4-DoF hand-held demonstrations to the 6-DoF wristed instruments of da Vinci-class robots.

Researcher (R&D Engineer)

Dec 2025 – present

MEDISC Lab, Prof. Hyoun-Joong Kong — Dept. of Transdisciplinary Medicine, Seoul National University Hospital

- Joined to automate laparoscopic surgery with Profs. Hyoun-Joong Kong and Kyu Eun Lee (Dept. of Surgery), in the line of SRT-H and SutureBot. Took the lab's idea of recording surgery with sensors on the surgeon's own instruments, rather than by teleoperation, from concept to the first instrument-mounted logger and the ex-vivo experiments; the lab spun this work out as Rosota.

Undergraduate Researcher

Sep 2025 – present

INRoL (Interactive & Networked Robotics Lab), Prof. Dongjun Lee — Dept. of Mechanical Engineering, Seoul National University

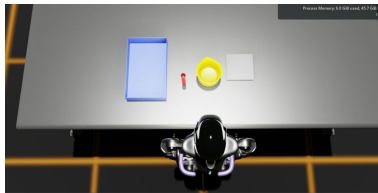
- **Humanoid robot foundation models.** Built an Isaac Sim to NVIDIA Isaac GR00T N1.5 bridge for the Fourier GR1, with fine-tuning and roll-out tutorials, pick-and-place and nut-pouring environments and an evaluation pipeline; documented it as the lab's "RFM Wiki" (Notion + GitHub), including what breaks: dataset jitter, fine-tuning budget, scene construction.
- **Teaching.** These tutorials were the hands-on material of the graduate course *Advanced Topics in Mechanical Design 1: AI Robotics and Robot Foundation Models* (Prof. Yongseok Lee, then at SNU, now DGIST), where I was the teaching assistant in Fall 2025.
- **Senior thesis: tactile simulation for humanoid hands** (with Somang Lee, Ph.D. student). Added a Pasternak-foundation tactile module to the lab's CRISP simulator so that stress propagates between neighboring taxels; benchmarked seven simulators $\times$ four indenters against FEniCSx FEM: peak-pressure error 4–17% at ≈ 0.5 ms per frame on a 60×60 grid. Now identifying the elastomer, friction and per-taxel parameters of a commercial fingertip sensor from robot indentation data with a differentiable forward model, for CRISP and Isaac Sim/Newton.

EARLIER

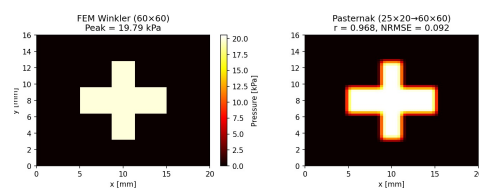
Founder & CEO, Lighter Links AI, Seoul, Aug 2022 – Aug 2025 (team of four). Shipped Clemory (RAG knowledge base), IsItYou (a chatbot that speaks as you; #3 Product of the Day on Product Hunt), Dabom Search API (Korean web data for LLM apps), Symphony (browser-based cloud desktop run by a vision agent) and ActualCode (open-source coding agent for hardware).

Research Intern, FriendliAI, Dec 2023 – Mar 2024 (Prof. Byung-Gon Chun). Low-level GPU programming for image-generation models.

Attack-Helicopter Mechanic, Republic of Korea Army, 2024 – Aug 2025. Maintenance crew for MD 500 Defender attack helicopters; completed the full eighteen-month service.



Fourier GR1 in Isaac Sim 5.0, nut-pouring task from my GR00T tutorial.



Tactile module: FEM ground truth (left) against my Pasternak model (right), cross indenter.



Closed-loop policy selection in a scene rebuilt from real organs.

PUBLICATIONS

* first or co-first author; † corresponding. [1]–[6] were submitted to IEEE ICRA 2027 in Sep 2026 and are under review.

- [1] D. Yee*, J. Oh, J. Lee, J. Lee, Y. Seo, S. Jeong, M. Kim, S. Shim, Y. Noh, H. Choi, Y. Kong, K. E. Lee†, H.-J. Kong†. *From Instrument-Mounted Demonstrations to In-Vivo Execution: Learning Bimanual Laparoscopic Appendectomy Without Robot-Collected Demonstrations*. [arXiv:2609.25625](#) · [dataset](#) · [video](#)
- [2] J. Lee*, D. Yee*, M. Kim, Y. Noh, S. Shim, J. Oh, Y. Seo, J. Lee, S. Jeong, H. Choi, H.-J. Kong†. *Drape-Compatible Tool-Tip Localization for Hand-Held Laparoscopic Instruments via UWB Carrier-Phase Ranging and Trocar-Constrained Geometry*.
- [3] J. Lee*, D. Yee, J. Oh, J. Lee, Y. Seo, et al., H.-J. Kong†. *Recording Hand-Held Laparoscopic Instrument Motion in the Operating Room: Magnetometer-Free Fusion of Inertial, Range and Visual Sensing*.
- [4] M. Kim*, S. Shim*, D. Yee, Y. Noh, J. Oh, et al., H.-J. Kong†. *Surgical Master Console Using General-Purpose Robot Arms and a Separable Articulated Distal Interface: Porcine In-Vivo Evaluation*.
- [5] J. Oh*, D. Yee, J. Lee, J. Lee, Y. Seo, et al., H.-J. Kong†. *Contact-Stable Deformable Tissue Simulation Using Implicit Integration and Live-Pose Grasp Constraints for Laparoscopic Surgery Robot Policy Evaluation*.
- [6] S. Jeong*, M. Kim, D. Yee, J. Oh, Y. Seo, et al., H.-J. Kong†. *Contact-Adaptive Robotic Ultrasound Probe Control for Tissue Exploration and Continuous Task-Relevant Visualization Using Robot-Free Image-Motion Demonstration*.
- [7] Y. Lee, H. Park, H. Kim, H. Ji, D. Yee, D. Lee. *Human-in-the-Loop Object Segmentation for 3D Gaussian Splatting via Finger-based VR Interface*. *Advanced Intelligent Systems* 8(8), e202501025, 2026. [doi:10.1002/aisy.202501025](#)

PATENTS

lead inventor on both; applicant Rosota Inc.

- *Detachable Surgical Data Recording Apparatus for Manual Laparoscopic Surgical Instrument*. KR patent application 10-2026-0150277 (filed 11 Aug 2026; priority 10-2026-0143776, 3 Aug 2026), 14 claims, examination requested. Inventors: D. Yee, Y. Seo, J. Oh.
- *Sensing and data-recording device for a laparoscopic instrument (partial design)*. KR design application 30-2026-0037733, filed 1 Oct 2026. Creators: D. Yee, Y. Seo, J. Oh.

EDUCATION

B.S., Mechanical Engineering and Computer Science & Engineering (double major)

Mar 2021 – Aug 2027 (expected)

Seoul National University — enrollment paused 2024–25 for military service; thesis in Prof. Dongjun Lee's INRoL

SKILLS

Robot learning	diffusion policy, VLA fine-tuning (GR00T), imitation learning, closed-loop evaluation in Isaac Sim and MuJoCo
Robotics & control	Franka FR3 with 1 kHz RCM control, teleoperation, error-state Kalman filtering, latency identification, FEM
Hardware	embedded firmware (nRF52840, ESP32; IMU, ToF, Hall, strain gauge, UWB), CAD (SolidWorks, Fusion 360)
Software	Python, C/C++, Rust, TypeScript; FastAPI, React, Docker, AWS; LLM systems (RAG, agents)
Languages	English and Korean, both native

Open to Ph.D. and collaboration conversations. Papers, videos and data: [rosota.ai](#); more at [donghoyee.github.io](#).